

# Syllabus - Beginner to Expert

## 🎓 SQL Syllabus – Beginner to Expert

### ● Level 1: Introduction & Basics

**Objective:** Understand databases, tables, and basic data manipulation.

| Week/Module | Topics                                                                                                                              |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1.1         | What is a Database and SQL?<br>Types of Databases (Relational vs. NoSQL)                                                            |
| 1.2         | Creating & Using a Database ( <code>CREATE DATABASE</code> , <code>USE</code> )                                                     |
| 1.3         | Creating Tables ( <code>CREATE TABLE</code> ) with Data Types ( <code>INT</code> , <code>VARCHAR</code> , <code>DATE</code> , etc.) |
| 1.4         | Inserting Data ( <code>INSERT INTO</code> )                                                                                         |
| 1.5         | Retrieving Data ( <code>SELECT</code> , <code>SELECT *</code> , <code>SELECT column_name</code> )                                   |
| 1.6         | Filtering Data ( <code>WHERE</code> , comparison operators)                                                                         |
| 1.7         | Sorting Results ( <code>ORDER BY ASC/DESC</code> )                                                                                  |
| 1.8         | Limiting Results ( <code>LIMIT</code> )                                                                                             |

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### ● Level 2: Intermediate Queries & Functions

**Objective:** Learn to update, delete, and summarize data.

| Week/Module | Topics                                                                                                                  |
|-------------|-------------------------------------------------------------------------------------------------------------------------|
| 2.1         | Updating Data ( <code>UPDATE</code> )                                                                                   |
| 2.2         | Deleting Data ( <code>DELETE</code> )                                                                                   |
| 2.3         | SQL Functions: <code>COUNT()</code> , <code>SUM()</code> , <code>AVG()</code> , <code>MIN()</code> , <code>MAX()</code> |

|     |                                                     |
|-----|-----------------------------------------------------|
| 2.4 | Grouping Data ( GROUP BY )                          |
| 2.5 | Filtering Groups ( HAVING )                         |
| 2.6 | Pattern Matching ( LIKE , % , _ )                   |
| 2.7 | Logical Operators ( AND , OR , NOT , IN , BETWEEN ) |
| 2.8 | Aliases ( AS ) and Column Renaming                  |

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### ● Level 3: Relationships and Joins

**Objective:** Understand table relationships and retrieve data from multiple tables.

| Week/Module | Topics                                                        |
|-------------|---------------------------------------------------------------|
| 3.1         | Primary Keys, Foreign Keys                                    |
| 3.2         | INNER JOIN                                                    |
| 3.3         | LEFT JOIN                                                     |
| 3.4         | RIGHT JOIN                                                    |
| 3.5         | FULL OUTER JOIN (conceptual; not directly supported in MySQL) |
| 3.6         | Self Join                                                     |
| 3.7         | Using ON vs USING                                             |
| 3.8         | Cross Join & Cartesian Product                                |

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### ● Level 4: Advanced SQL Techniques

**Objective:** Perform complex queries, subqueries, and use set operations.

| Week/Module | Topics                                             |
|-------------|----------------------------------------------------|
| 4.1         | Subqueries (Scalar, Column, Row, Table subqueries) |
| 4.2         | Correlated Subqueries                              |

|     |                                                                                  |
|-----|----------------------------------------------------------------------------------|
| 4.3 | EXISTS , NOT EXISTS                                                              |
| 4.4 | Set Operations: UNION , UNION ALL , INTERSECT , EXCEPT                           |
| 4.5 | CASE Statements and Conditional Logic                                            |
| 4.6 | Derived Tables (Inline Views)                                                    |
| 4.7 | Views: CREATE VIEW , ALTER VIEW , DROP VIEW                                      |
| 4.8 | Window Functions (if using MySQL 8+): ROW_NUMBER() , RANK() , OVER(PARTITION BY) |

### ● Level 5: Schema Design & Optimization

**Objective:** Understand schema design, constraints, indexing, and performance.

| Week/Module | Topics                                                            |
|-------------|-------------------------------------------------------------------|
| 5.1         | Constraints: NOT NULL , UNIQUE , DEFAULT , CHECK , AUTO_INCREMENT |
| 5.2         | Composite Keys & Surrogate Keys                                   |
| 5.3         | Indexes: CREATE INDEX , DROP INDEX , pros/cons                    |
| 5.4         | Transaction Control: BEGIN , COMMIT , ROLLBACK , SAVEPOINT        |
| 5.5         | Isolation Levels (Read Uncommitted to Serializable)               |
| 5.6         | Stored Procedures (Intro only)                                    |
| 5.7         | Triggers (Intro only)                                             |
| 5.8         | Query Optimization Basics (EXPLAIN, execution plans)              |

### ■ Optional Topics (For Advanced Learners)

- JSON support in MySQL
- Full-text search

- CTEs (Common Table Expressions)
- Temporal tables
- Partitioning

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